

Einführung in die Kern-, Teilchen und Astrophysik

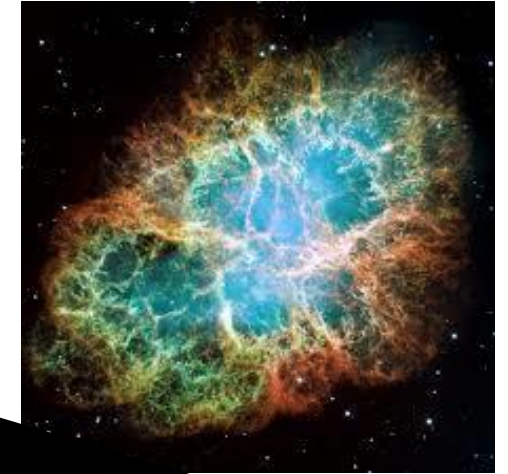
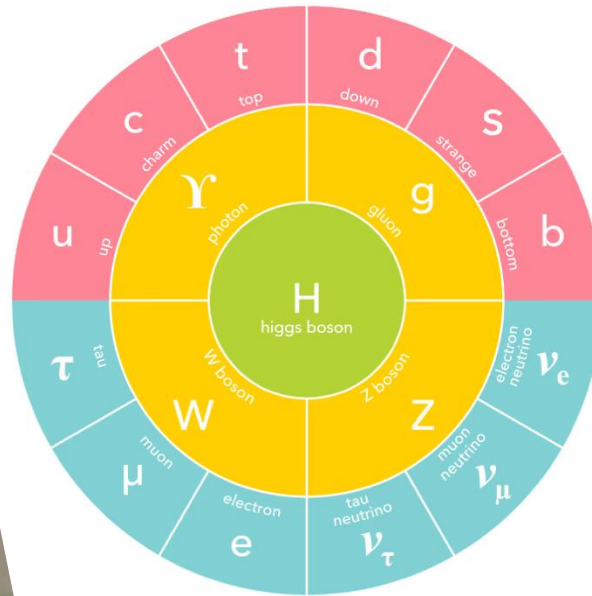
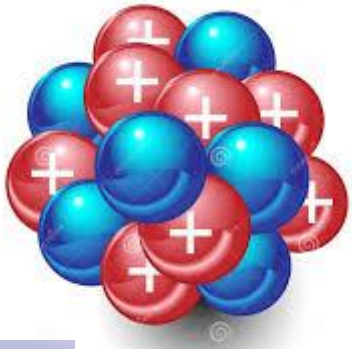


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Lecture	Date	Theme	Topic	Sub-Topics
1	14/10/25	Introduction	Little History of particle physics	atoms, elementary particles
2	21/10/25	Particle Physics	Standard Model	Fermions, Bosons, Interactions,
3	28/10/25		Fundamental symmetries	Conservations, Feynman, Propagator,
4	04/11/25		Fundamental symmetries	Discrete symmetries; Cross section
5	11/11/25		Interaction with Matter	Bethe Bloch, gamma interaction
6	18/11/25		Particle detection	Gamma interactions, detector basics
7	25/11/25	Astrophysics	Detectors, accelerators	Detector types, accelerators, cyclotron Supernovae, cosmic rays, Supernovae, cosmic rays
8	02/12/25	Nuclear physics	Basics	Binding energies, Weizäcker
9	09/12/25		Nuclear reactions	Beta, alpha, fission
10	16/12/25		Nuclear reactors	Radioactive transitions, Radiocarbon method
11	13/01/26	Particle / Nuclear	Nuclear Structure	Gamma transitions, Mott, Form factors
12	20/01/26		Nucleon Structure	Deep inelastic scattering, Quarks and QCD
13	27/01/26	Particle Physics	Weak interactions	
14	03/02/26	Astrophysics	Neutrinos	

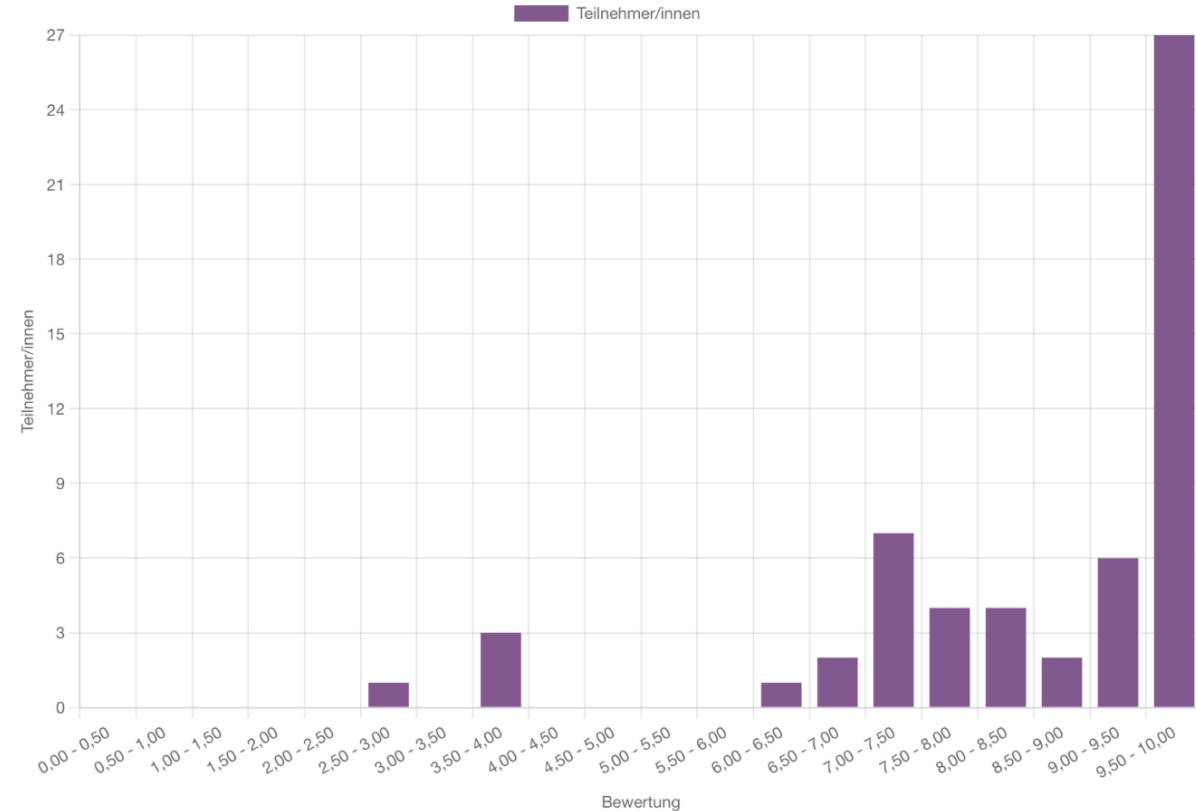


Quiz am 23.12.25 --- Online in Moodle

- Bestandenes Quiz als Voraussetzung für Notenbonus

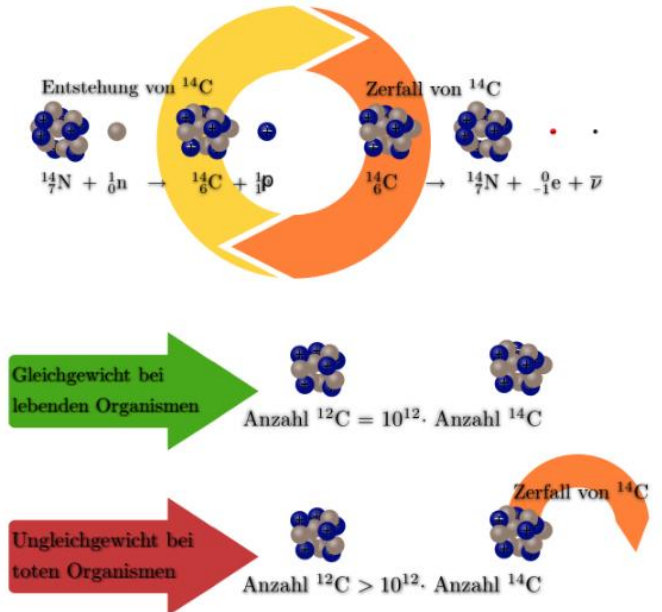
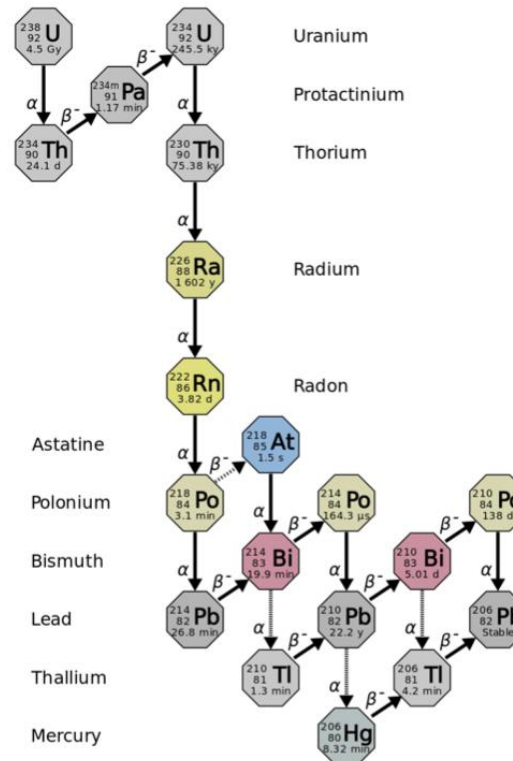
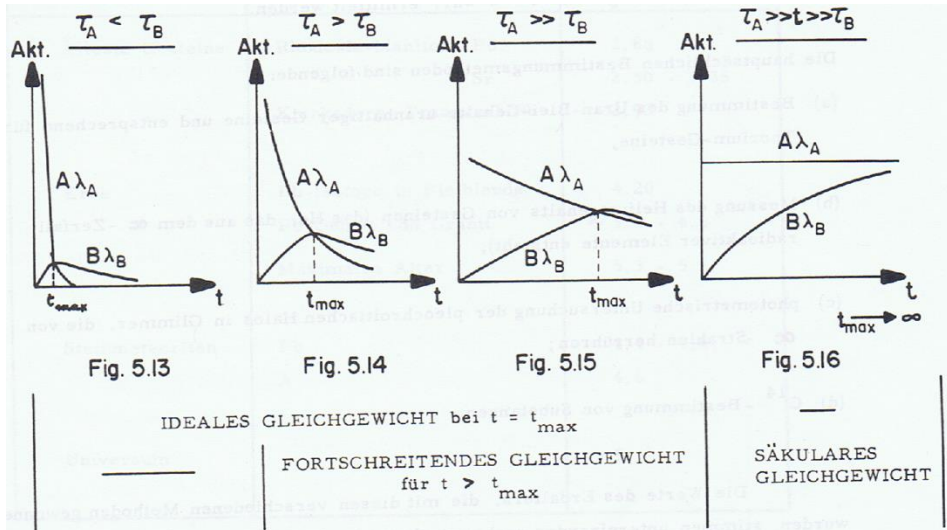


Prüfung: 26.2. at 11:00 -12:30



Zusammenfassung der letzten Vorlesung

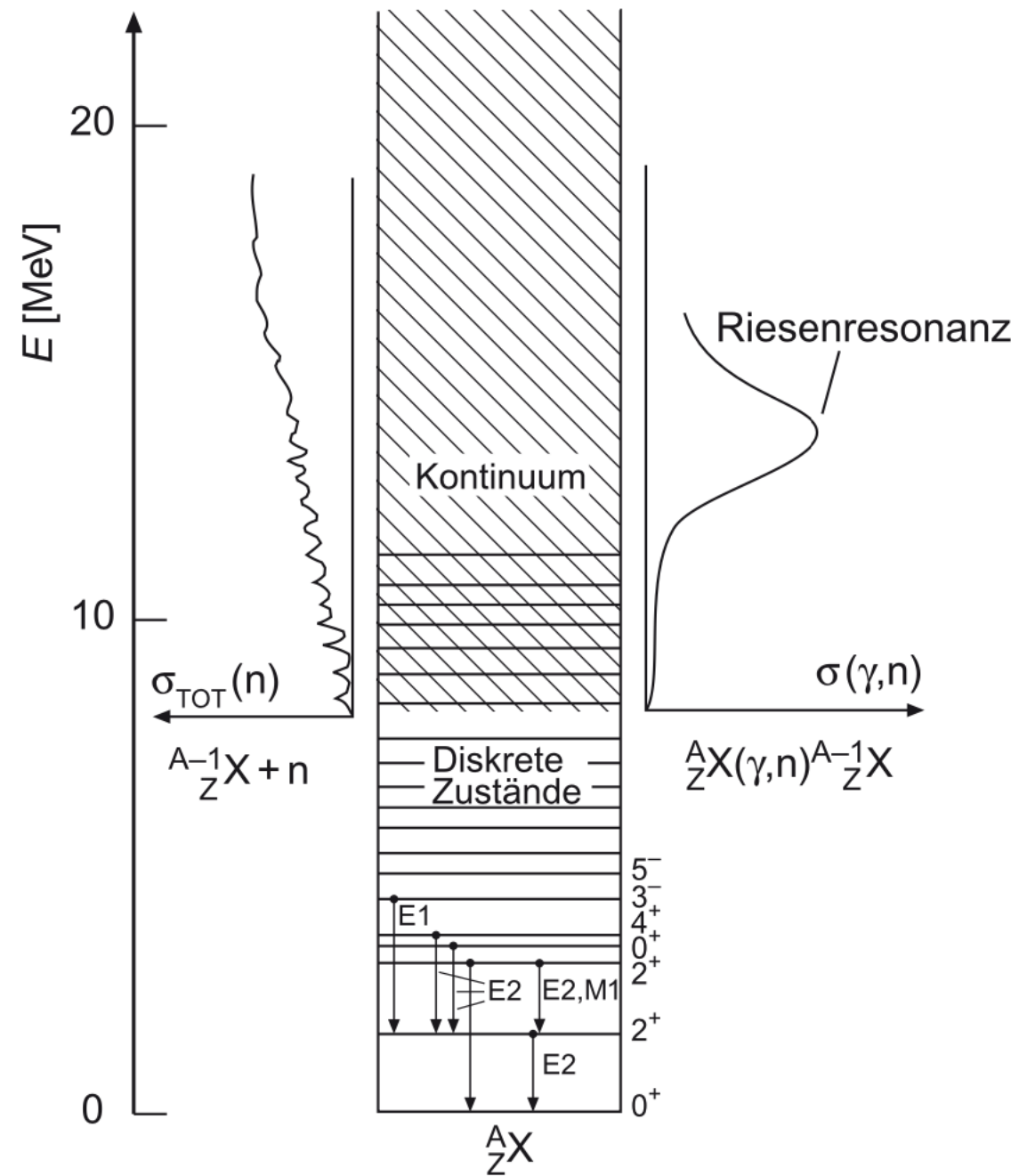
$$A = N \cdot \lambda = -\frac{dN}{dt}$$

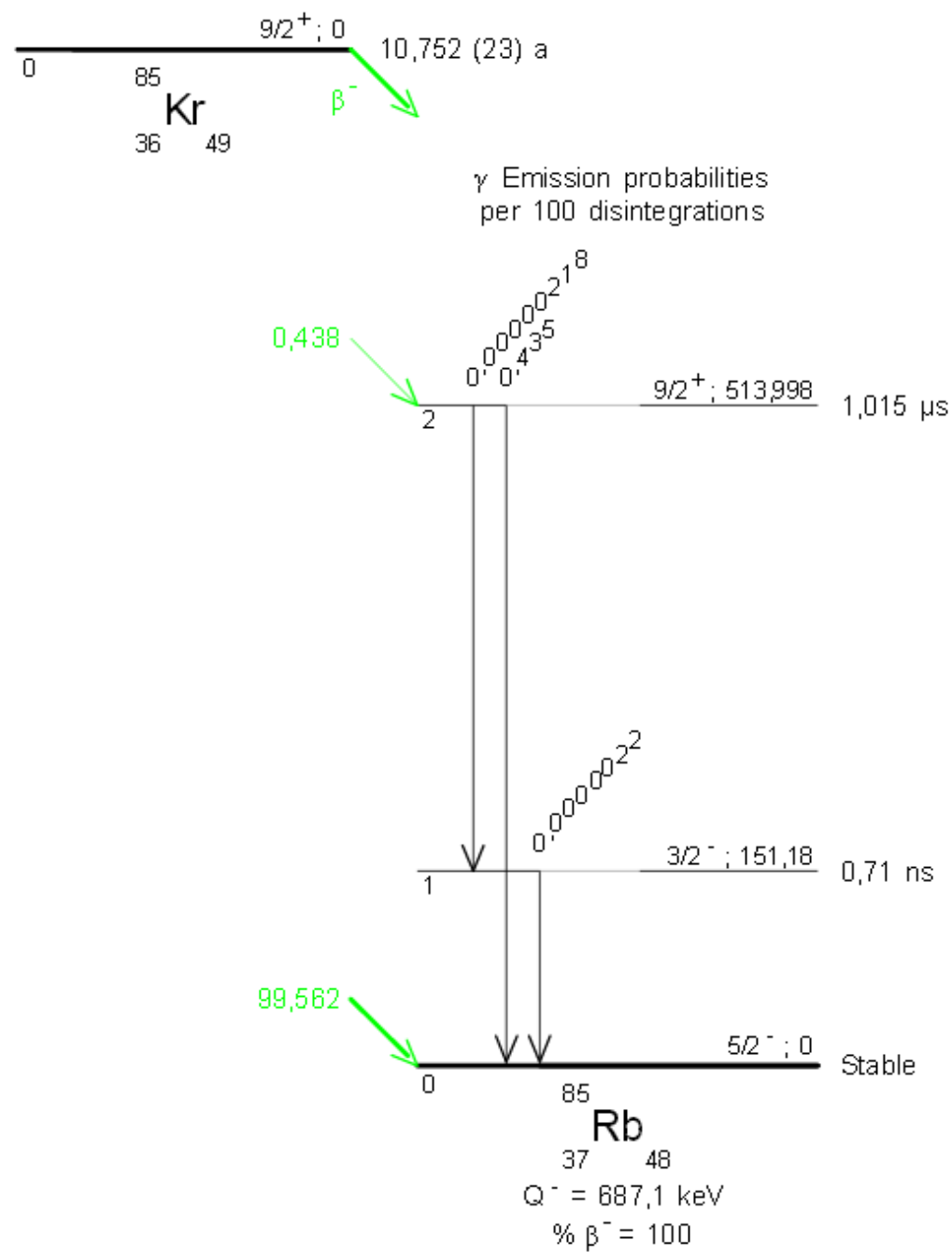


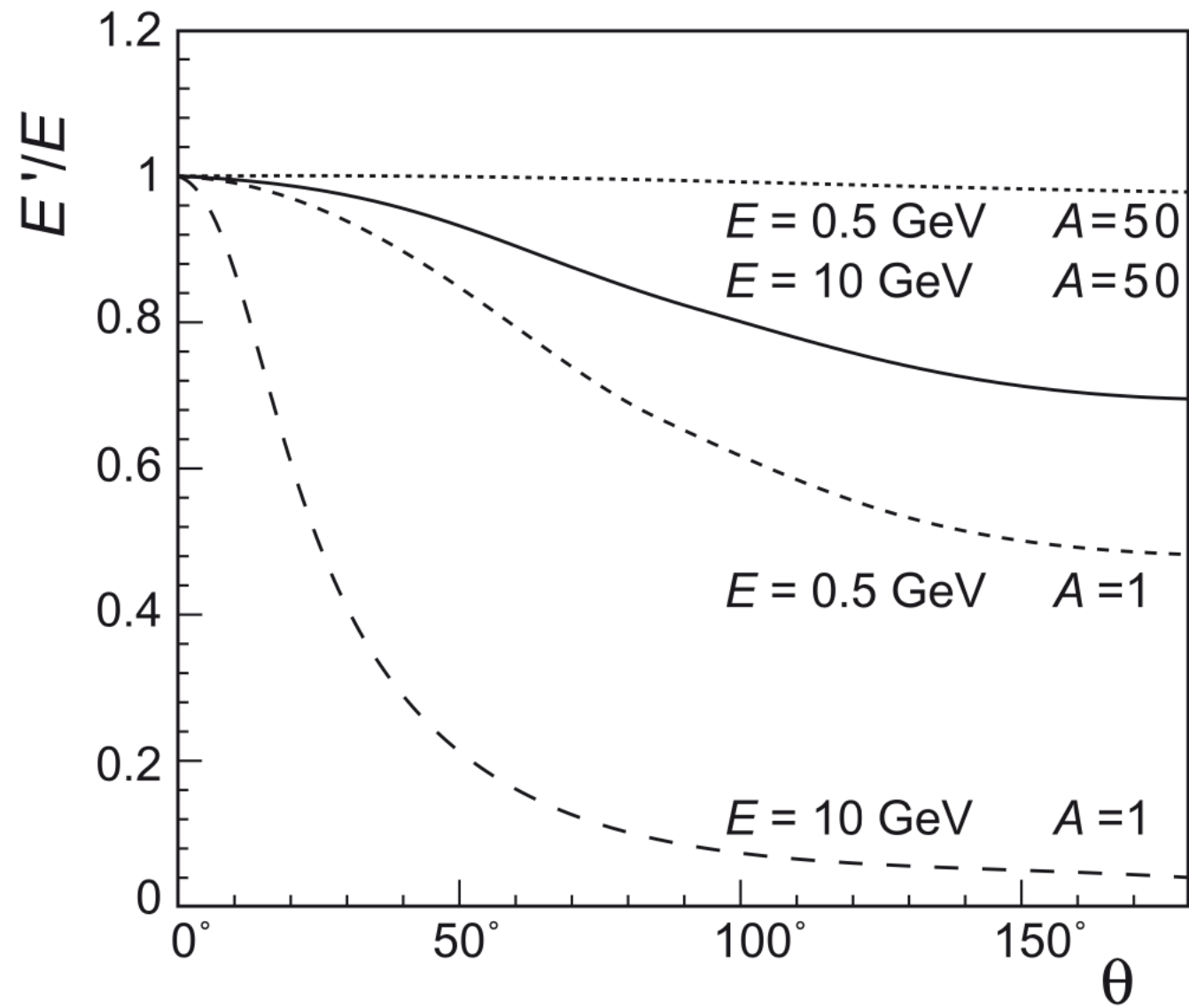
Aktivität in radioaktiven
Zerfällen

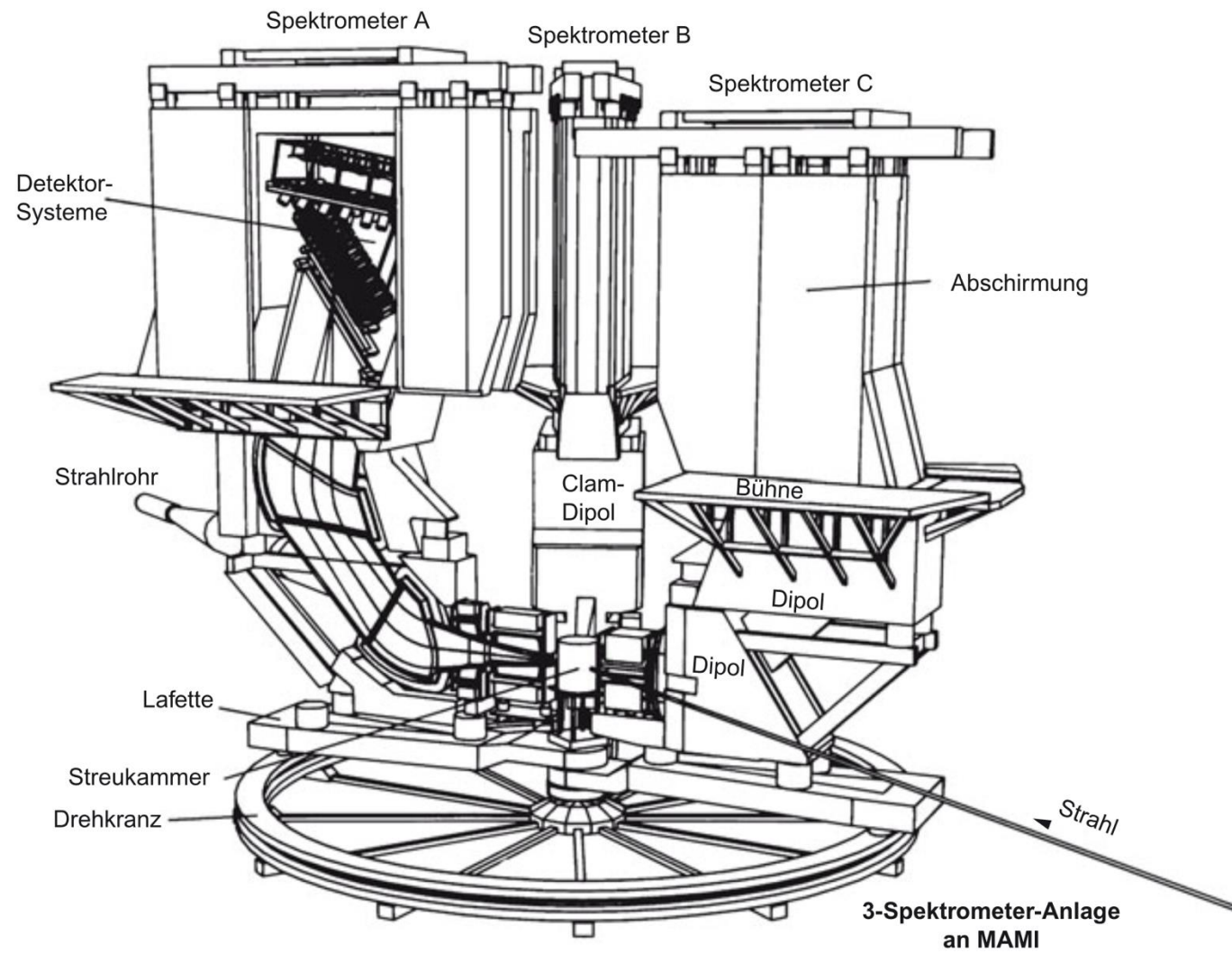
Zerfallsketten

Radiokarbon-
methode



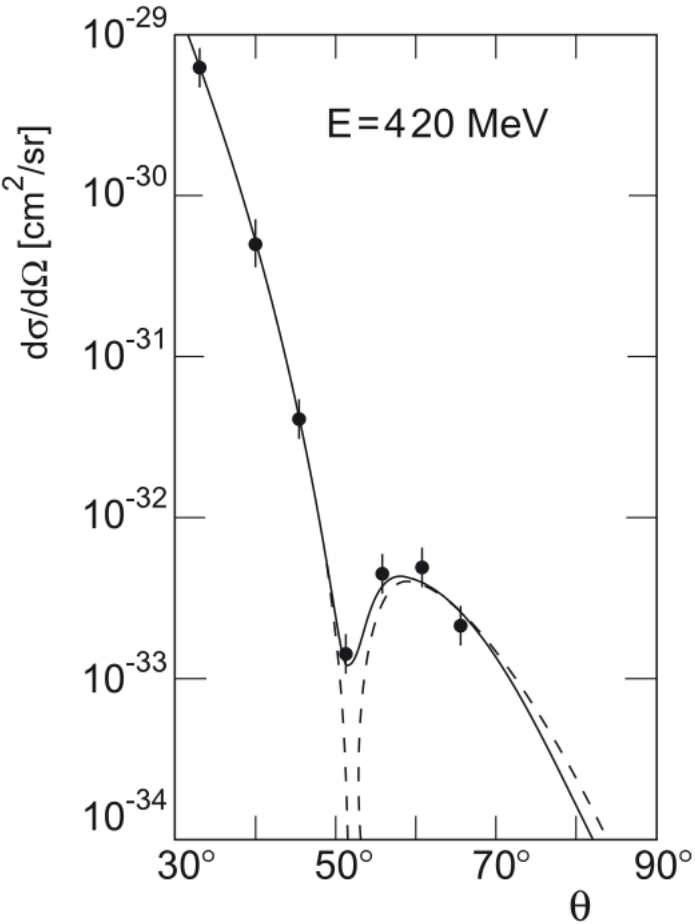




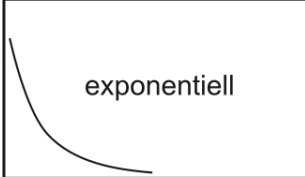
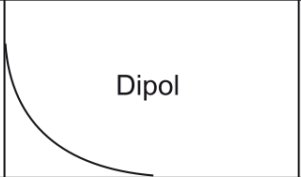
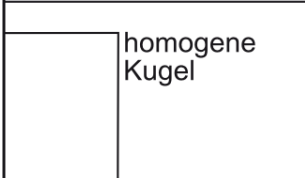
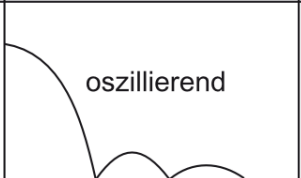
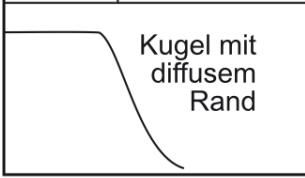
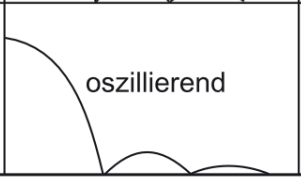


Povh Abb 5.4

Elektron-Streuung an ¹²C



Povh Kap. 5

$\rho(r)$	$ F(\mathbf{q}^2) $	Beispiel
 punktförmig	 konstant	Elektron
 exponentiell	 Dipol	Proton
 gaußförmig	 gaußförmig	⁶ Li
 homogene Kugel	 oszillierend	—
 Kugel mit diffusem Rand	 oszillierend	⁴⁰ Ca
$r \longrightarrow$		$ \mathbf{q} \longrightarrow$

Ladungsverteilung $f(r)$		Formfaktor $F(\mathbf{q}^2)$	
Punkt	$\delta(r)/4\pi$	1	konstant
exponentiell	$(a^3/8\pi) \cdot \exp(-ar)$	$(1 + \mathbf{q}^2/a^2\hbar^2)^{-2}$	Dipol
Gauß	$(a^2/2\pi)^{3/2} \cdot \exp(-a^2r^2/2)$	$\exp(-\mathbf{q}^2/2a^2\hbar^2)$	Gauß
homogene Kugel	$\begin{cases} 3/4\pi R^3 & \text{für } r \leq R \\ 0 & \text{für } r > R \end{cases}$	$3\alpha^{-3}(\sin\alpha - \alpha\cos\alpha)$ mit $\alpha = \mathbf{q} R/\hbar$	oszillierend

